engineering, we will develop a web application that can answer queries about the data we provide. The application will use techniques like grounding to ensure the generated responses remain faithful to the source documents, enhancing the accuracy and reliability of the AI-generated content.

You should be comfortable with the Vercel AI SDK (chapters 3–4) and LangChain.js workflows (chapter 6). Familiarity with prompt engineering techniques (chapter 5) will help you implement effective summarization and RAG strategies.

7.1 Building a document summarization web application with LangChain.js

We've used the Vercel AI SDK to create reactive UIs powered by large language models (LLMs), and LangChain.js to develop scalable conversational platforms capable of complex data processing. Now we will use that know-how to build a web application that uses generative AI to take complex documents and produce concise, meaningful summaries. This app features a user-friendly web interface that accepts user-submitted documents, such as Word (DOCX) files or PDFs. Some of the challenges we'll face include addressing inaccuracies, managing large documents that exceed the LLM's context window, and ensuring the quality of summaries.

7.1.1 Summarization app project requirements

The application we are going to build has these key elements:

- *Web interface for document submission*—A user-friendly interface that allows users to upload documents, such as Word files or PDFs, for summarization.
- *Document-processing pipeline*—An architecture that includes steps for document loading, text extraction, chunking, and semantic embedding to prepare the content for summarization.
- *Summarization workflow*—Implementation of various summarization techniques using LangChain.js and LLMs, enabling the generation of concise summaries from the processed documents.
- *Integration with the Vercel AI SDK*—We will utilize server actions to provide a nice user experience while summarizing documents.

7.1.2 Architecture and workflow

Before writing any code, we want to establish the components and technologies that will bring this web application to life. This overview will help us define the scope and determine which libraries and tools to use.

In this document summarization service, the web application architecture follows a pipeline approach. Each document moves through various processing stages from upload to summary generation, creating a streamlined workflow. We'll use a set of familiar technologies that we've already explored: